



Fiscal deficits, financial fragility, and the effectiveness of government policies[☆]



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ABSTRACT

Recent developments in the euro area highlighted the interactions between fiscal policy, sovereign debt and financial fragility. We introduce asset choice and sovereign debt holdings in banks' portfolios in an otherwise standard macroeconomic model with financial frictions, to emphasize a new crowding-out mechanism through reduced private access to credit when leverage-constrained banks accumulate sovereign debt. When banks are substantially invested in sovereign debt, the effectiveness of fiscal stimuli is impaired because deficit-financed fiscal expansions through this channel crowd out private demand. This channel also significantly reduces the gains from fiscal policy when interest rates are at the Zero Lower Bound.

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1. Introduction

Recent macro-developments in the euro area have highlighted the interactions of fiscal policy, sovereign debt, and financial fragility. Financial fragility and spiralling sovereign debt were inherited from the global credit crisis that preceded the more recent turmoil in the euro area. Across much of Europe and in the U.S., the fiscal response to the crisis took the form of financial sector support measures and economic stimulus packages that were financed through budgetary deficits. What are the effects of such policies in a situation of financial distress such as emerged after the 2007/2008 credit crisis? Standard macroeconomic models are not set up for policy analysis during crises characterized by financial distress, but new types of models have been developed recently that can be (and have been) used to answer this question; see, in particular, Gertler and Karadi (2011), Gertler and Kiyotaki (2010), and Gertler et al. (2012). However, those models assume that government policies are fully funded through markets that are not affected by financial distress and in turn have no direct impact on the financial fragility that the distress situation has led to. For many, if not most countries, this is not a realistic assumption. This paper shows that relaxing that assumption qualitatively changes the assessment of the effectiveness of deficit-financed government policies as a tool to fight recessions caused by financial distress, and underscores the importance of bank intervention policies to relieve financial distress.

We develop a structural macroeconomic model that integrates government deficit financing and financial intermediation with frictions in the intermediation process. These frictions imply a financial accelerator that is able to generate a deep

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financial crisis following a deterioration of intermediary balance sheets. Furthermore, we incorporate one of the more contentious points of the recent euro area crisis, the potential consequences of substantial holdings of sovereign debt by commercial banks for private access to bank credit. The model is used to study the effects of deficit-financed fiscal stimulus policies during such a crisis and also analyze the macroeconomic impact of policies designed to reduce financial distress through direct intervention in commercial bank balance sheets.

Our framework includes financial intermediaries that channel funds or deposits from households (the saving agents) to non-financial firms and the government (the borrowing agents). The intermediation process is subject to an agency problem between depositors and intermediaries, like in [Gertler and Karadi \(2011\)](#) and [Gertler and Kiyotaki \(2010\)](#). As those studies show, this agency problem leads to endogenous leverage constraints, which in turn gives rise to a powerful financial accelerator mechanism. This accelerator mechanism generates dynamics that broadly reflect the relevant economic dynamics of a financial crisis. However, unlike the models in those studies, our framework allows for different classes of assets in intermediary portfolios instead of only one class, introducing a new crowding-out channel that works through reduced private access to credit when banks accumulate sovereign debt under a leverage constraint.

The specific setup is as follows. There are two types of financial intermediaries, banks and Money Management Funds (MMFs). The asset portfolio of banks consists of government bonds and loans made to non-financial firms. The overall portfolio size of banks is tied to their equity capital through endogenous leverage constraints, but the banks optimize their portfolio composition for any given portfolio size, shifting the composition of their portfolios towards assets with higher expected returns. Through this mechanism, the expected returns on bonds and private claims are jointly determined in equilibrium. Financial frictions and leverage constraints prevent perfect arbitrage, although intermediaries will alter their portfolio composition to exploit rate of return differences. In general equilibrium, such arbitrage behavior by banks leads to co-movements between different credit spreads relative to the rates at which the intermediaries obtain funding.

MMFs do not engage in risky lending, they solely invest in government bonds which can be used as collateral. Therefore, MMFs are not subject to leverage constraints. The government decides on which share of its stock of debt is placed at banks and which share at MMFs. Placement at MMFs is equivalent to direct holdings by consumers as MMFs are modeled as zero-cost/zero-profit institutions.

We use this model to highlight the links between government policies and deficit financing in a situation of financial distress, analyzing in particular the effects of demand (i.e. spending) stimulus and measures targeting the financial sector (i.e. transfers to intermediaries). This set of policies is sufficient to explain the key implications of the model, but it also suitably captures the main fiscal and bank intervention policy measures that were applied in the wake of the global credit crisis. The policies are financed by issuing bonds to intermediaries. We analyze as a benchmark the case where governments can bypass financial frictions altogether by financing policies exclusively through MMFs and explore the consequences of governments having to finance an increasing share of debt through institutions subject to financial frictions. Key to our results are the effects of government borrowing on intermediary balance sheet constraints and the associated adverse impacts on the cost of credit to non-financial firms. In our model, a fiscal expansion is associated with an economy-wide increase in credit spreads for the private sector, as higher government deficits tighten intermediary balance sheet constraints. The rise in spreads lowers non-financial sector investment, which can (in some cases more than) offset the output gain of a demand stimulus. The fact that intermediary balance sheet constraints are forward-looking explains the sometimes perverse links between the timing and the effects of government policies that we detect.

Our findings suggest that intermediary financing under leverage constraints has important consequences for the effectiveness of government policies. An early, immediately implemented demand stimulus dampens a recession due to a financial crisis for some time (i.e. it reduces output losses initially), but the dampening impact is much smaller than without the leverage constraints and the associated crowding out through bank portfolio's. And the stimulus tends to prolong the downturn later on as within-period multipliers turn negative already after a few periods. An even more striking result is that a pre-announced future stimulus starts out by worsening the recession: in the year leading up to the actual implementation of the stimulus, within-period multipliers are actually negative. Since in the vast majority of countries budget procedures imply time to implementation, with delays of at least a year if not more typical, this is a highly relevant result. We also find that the crowding-out effects of deficit-financed fiscal expansions significantly limit the gains from fiscal policy when monetary policy is restricted by the Zero Lower Bound. Overall, these findings raise questions about the effectiveness of deficit-financed government policies in situations of financial distress and limited direct capital market access by governments.

This paper is part of a recent literature documenting the large holdings of sovereign bonds by banks ([Gennaioli et al., 2014a](#); [van der Kwaak and van Wijnbergen, 2014](#)) and studying their implications for whether sovereign default happens and what are its consequences ([Balloch, 2016](#); [Broner et al., 2014](#); [Gennaioli et al., 2014b](#)). We also highlight the importance of the links between financial fragility and bank holdings of sovereign debt, but our focus is different, on their impact on the size of the fiscal multiplier. This paper is also closely related to other studies of the credit crisis (e.g., [Christiano and Ikeda, 2013](#); [Gertler and Karadi, 2011](#); [Gertler and Kiyotaki, 2010](#); [Gertler et al., 2012](#)); but in these studies, government policies are financed directly by households without any financial frictions. However, many financial institutions are active in government funding markets. For instance, in the euro area, government securities and direct loans to the government equal to

about 58% of the value of loans to non-financial corporations,¹ so our highlighting of the importance of bank holdings of sovereign debt has empirical relevance.

The remainder of the paper is structured as follows. Section 2 briefly reviews the related empirical literature to connect the key mechanisms and predictions of the model to the available evidence. Section 3 lays out the model. Section 4 discusses the results of model-based simulations. It first compares the effects of deficit-financed changes in government purchases in comparison to a baseline model without financial intermediation to explain the main mechanisms at play. It then analyzes the effects of alternative fiscal policy responses to a simulated crisis to assess the stabilization properties of different policies, focusing also on the effectiveness of fiscal policy at the Zero Lower Bound. Section 5 concludes.

2. Related empirical evidence

This section provides a brief review of the empirical evidence that is relevant to our study. The related evidence can be grouped into (a) results from fiscal VAR studies on the effects of government spending or goods purchases, in particular on private investment, (b) findings of cross-sectional studies that investigate the impact of financial sector policies and fiscal stimulus during financial crises, and (c) results from empirical studies looking at the effects of fiscal finances on interest rates.

2.1. Fiscal VAR studies

Among the first group of studies, both structural VAR (SVAR) methods and event-study approaches point towards negative effects of government spending on private investment. On the SVAR side, Blanchard and Perotti (2002) find that investment is consistently crowded out by government spending shocks in the U.S. over the period 1960Q1–1997Q4, with a peak decline of up to 1% due to a 1% of GDP spending increase. Using a yearly panel VAR on 18 OECD countries over the period 1960–1996, Alesina et al. (2002) also find a sizable negative effect of public spending (particularly public wages) on investment, a one percentage point increase in the primary spending-to-GDP ratio leading to a fall in the investment-to-GDP ratio of 0.15 percentage points on impact and to a cumulative fall of 0.74 percentage points after five years.

On the side of event studies, identifying spending shocks based on war dates and professional forecasts, Ramey (2011) finds that after a positive defense news shock in the U.S. both non-residential and residential investment fall significantly, with peak effects of up to –1% (non-residential investment) and –1.5% (residential investment). Shocks identified based on professional forecast errors over the period 1969–2008 indicate even stronger falls of –1.5% and –3.5%, respectively, as well as a medium-term decline in output.

The model results we will present are in the direction of the VAR studies mentioned in the previous paragraphs although in our model the negative effects are somewhat stronger than those documented in the empirical studies reviewed above (see Figs. 1 and 3).² The quantitative differences should not come as a big surprise as our model mainly describes business cycles in times of financial distress, whereas the above studies look instead at sample averages also including “normal” times. So the results of cross-sectional studies, which we discuss next, are possibly more relevant.

2.2. Cross-sectional studies

Mulas-Granados et al. (2009) estimate the effects of fiscal policy interventions during 118 episodes of banking crises in a cross-section of developed and emerging countries. In line with our results of Section 4.5, they find that financial sector policies can shorten such crises whereas fiscal stimulus going along with such policies can have stabilizing effects, but not for countries where fiscal policy is subject to funding constraints. A similar conclusion is reached by Ilzetzki et al. (2013); based on a quarterly dataset for 20 developed and 24 developing countries over the period 1960–2007, they find that during episodes where government debt was higher than 60% of GDP, spending multipliers are not statistically different from zero on impact and negative (and statistically significant) in the long run.

2.3. Effects of fiscal finances on interest rates

A recent study on the effects of fiscal finances on interest rates has been conducted by Laubach (2009), who estimates the effects of U.S. government debt and deficits on Treasury yields, isolating those effects from other factors affecting interest rates (such as countercyclical monetary policy and automatic fiscal stabilizers) by focusing on the relation between long-horizon expectations of both interest rates and fiscal variables. According to Laubach (2009), the idea is that measures of expectations hold out the prospect of uncovering causal effects from fiscal variables to interest rates. Laubach (2009) concludes that the effects of fiscal variables on interest rates are statistically significant and economically relevant; in

¹ Source: European Central Bank (ECB), aggregated balance sheet of euro area monetary financial institutions, March 2011. Updated versions are available on the ECB's webpage: <http://www.ecb.int/stats/money/aggregates/bsheets>

² The fall in investment in Fig. 1 implies a 0.4 percentage points decline in the investment-to-GDP ratio on impact and a cumulative fall of approximately four percentage points after five years.

particular, for the period 1976–2006 an increase in the projected deficit-to-GDP ratio by one percentage point raises forward rates five and more years into the future by about 25 basis points. This result is in line with our findings shown in Figs. 1 and 3. Other related studies also tend to find statistically and economically significant effects of deficits on interest rates (see Canzoneri et al., 2002; Elmendorf, 1993; Wachtel and Young, 1987).

2.4. Other studies

Finally, a new view on the size of fiscal multipliers is presented in Blanchard and Leigh (2013), see also IMF (2012). Blanchard and Leigh (2013) derive an estimate of the “true” multiplier from the forecast errors made by IMF staff in 2010 using the actual outcomes in a wide range of crisis countries and conclude that true multipliers must on average equal 1.5 and in the range of 0.9–1.7 to explain the forecast errors, substantially higher than assumed in the 2010 forecasting exercises by almost a full point. However, the results are based on an ad hoc assumption on the multiplier used in those 2010 forecasting exercises; in particular they assume that the multiplier used in the first benchmark half of the period they consider was also used in the second half where the forecast errors occurred that lead them to conclude the multiplier must have been higher. Górnicka et al. (2014) repeat this exercise for European crisis countries using actual information on the forecasts and no-consolidation counterfactuals used, obtained from early Excessive Deficit Procedures instigated by the European Commission. They show that the multiplier assumptions used actually changed over time, and that incorporating that information leads to substantially lower estimates of the average forecast errors (0.5 instead of the Blanchard–Leigh assumption of 1.0), with correspondingly lower estimates for the actual multiplier, an average below 1.0 compared to the Blanchard–Leigh average of 1.5. These estimates are more in line with the results in this paper. Moreover, although both Górnicka et al. (2014) and Blanchard and Leigh (2013) control for crisis severity, neither controls for whether a recapitalization of banks was attempted. The point of our paper is that in countries with adequately recapitalized banks multipliers would have been higher. Investigating that prediction is on our agenda of future work.

3. Model description

We describe a monetary model with sticky prices and financial intermediation that builds on Christiano et al. (2005) and Gertler and Karadi (2011) but extends the basic structure significantly to enable analysis of commercial banks' role in public deficit financing. This section describes the model structure, with the main emphasis on the key equations for the financial institutions that play a role in our setup and the new assumptions that we introduce. The model has a private sector and a public sector. The private sector consists of a non-financial sector that is formed by households and firms, and a financial sector that is formed by two types of financial intermediaries, banks and Money Management Funds (MMFs). The public sector is formed by a monetary authority that sets the risk-free nominal interest rate and a government that conducts purchases of the final good and implements financial sector policies. The government finances its operations by issuing debt to banks that are subject to financial frictions and to MMFs that are not. The formal structure of the model, description of the steady state solution and the dynamics are listed and discussed fully in the appendix.

3.1. Standard non-financial structure

Consider the non-financial sector first. This model segment is by now standard, so the detailed equations describing all this are relegated to the appendix. The firm production chain is as follows. Capital producers combine used capital purchased from intermediate goods producers with investment goods to produce new productive capital that is again purchased by intermediate goods producers. The latter rent labor services from households, issue claims to banks to finance their capital acquisition, and produce differentiated goods that are bought, re-packaged and sold by retail firms in a monopolistically competitive market. Final goods producers buy those goods and combine them into a single output good.

There is a continuum of infinitely lived households with identical preferences and identical asset endowments. Following Gertler and Karadi (2011), within each household there is a fraction $1 - \zeta$ of workers that supply labor to firms and a fraction ζ of bankers that operate financial intermediaries (banks). There is perfect consumption insurance within the family. Households save by holding deposits at intermediaries that they do not own, either at banks or at Money Management Funds. Bankers operating financial intermediaries have finite life times, to exclude self-financing equilibria. Thus, at the beginning of each period, with probability $1 - \theta$ an individual intermediary/banker exits and with probability θ the intermediary/banker continues operating. All profits are retained as capital, there are no dividends. If the intermediary exits, the respective bankers become workers and transfer all retained capital back to the household which owns that intermediary. Thus every period $(1 - \theta)\zeta$ bankers become workers. To keep the relative proportions fixed, a similar number of workers become bankers. New bankers receive a start-up transfer from their household.

Household preferences depend on consumption and labor supply, with habit formation in consumption as in Christiano et al. (2005) to capture consumption dynamics. The objective of a representative household in period t is to maximize

expected discounted utility

$$\mathcal{W}_t = E_t \sum_{s=0}^{\infty} \beta^s [\log(c_{t+s} - v c_{t-1+s}) - (1+\varphi)^{-1} h_{t+s}^{1+\varphi}], \quad (1)$$

with $\beta \in (0, 1)$, $v \in [0, 1)$ and $\varphi > 0$, subject to the period-by-period budget constraint $c_t + d_t + \tau_t \leq w_t h_t + (1+r_t^d) d_{t-1} + \Sigma_t$, where c_t denotes consumption of final goods, h_t denotes hours worked, w_t is the hourly wage rate, d_{t-1} are beginning-of-period deposits, d_t are end-of-period deposits, r_t^d is the net real interest rates on deposits, τ_t are lump-sum tax payments, and Σ_t collects payouts from ownership of both non-financial and financial firms, net of transfers given to household members that enter as bankers at time t . The household's decision problem is subject to a no-Ponzi game condition, and the household takes w_t , r_t^d , τ_t , Σ_t , prices, and its initial wealth endowment d_{-1} as given. The solution to the ensuing constrained maximization problem is standard and relegated to the appendix.

3.2. Financial intermediaries: money management funds

Banks are subject to informational frictions, like in [Gertler and Karadi \(2011\)](#), Money Management Funds (MMFs) are not. The reason for introducing a second frictionless intermediary (which by assumption is fully invested in government bonds) is to incorporate the notion that informational frictions are a smaller problem for government bonds than for corporate paper. In a Holmstrom–Tirole world, one could simply assume that a higher fraction of their future returns are “pledgeable”, hence a smaller part has to be financed through equity, but that is more difficult to incorporate in the [Gertler and Karadi \(2011\)](#) setup we use in this paper. Hence the assumption that a fraction of new bond issues is allocated to banks subject to informational frictions, and the remaining fraction to frictionless MMFs. This setup closely mimics the parallel institutions through which sovereign debt is held in the U.S. and in Europe; while MMFs are a uniquely U.S. institution, funds that are fully invested in sovereign debt are widely accessible in Europe and fulfill a similar function. Bond holdings by commercial banks are likely to be more important as a share of total debt outstanding in countries where capital markets are less developed, although regulatory pressure has also led to very large volumes of government debt in bank portfolios, particularly in Western Europe. On the other hand, U.S. banks play a smaller role in intermediating public debt. To reflect that variety we introduce a second intermediary specializing in government debt only, Money Management Funds. MMFs are zero cost intermediaries not subject to any financial frictions. The option to exclusively finance through MMFs, thus bypassing financial frictions altogether, is introduced as a benchmark only, to gauge the impact of the interaction between financial frictions and the financing of government expenditure.

Both intermediaries take funds from depositors that are remunerated at the risk-free nominal interest rate. The deposits in MMFs are considered perfect substitutes for bank deposits as long as banks satisfy the incentive compatibility constraint elaborated upon below. The need for the government to resort to intermediaries instead of placing debt directly with consumers/depositors is motivated by size arguments: bond issuance typically occurs in large tranches that cannot be handled by small investors. A fraction Δ_t of all new issues is allocated to banks, and the remaining $1 - \Delta_t$ is allocated to MMFs. Both MMFs and banks hold on to government paper until they mature (all bonds are short-term one-period bonds).

MMFs take private deposits and are fully invested in government bonds. Consumers consider bank deposits and MMF deposits as perfect substitutes. Bonds placed with MMFs thus have to earn a return equal to the deposit rate. MMFs are simply pass-through windows for the share of government bonds allocated to them and face no incentive compatibility constraint. Since they face no informational frictions, they are completely financed by issuing deposits. Their assets p_t^{MMF} are equal to the share of government bonds allocated to them, financed by taking in deposits d_t^{MMF} . Their aggregate balance sheet constraint is thus simply $p_t^{MMF} = d_t^{MMF}$. We assume that MMFs are zero-cost, zero-profit pass-through windows for government bonds, so MMFs get paid $r_t^{b,MMF} = r_t^d$ on government bonds. As long as the leverage constraint binds and deposits in banks and MMFs are perfect substitutes in consumer portfolios, this implies that government bonds have to be sold at a discount to bank intermediaries.

3.3. Financial intermediaries: banks

Deposit banks are competitive and located on a continuum indexed by $j \in [0, 1]$. The bank intermediaries use the deposits obtained from households to purchase claims issued by intermediate goods firms and also government bonds. Each bank intermediary chooses its asset holdings to maximize the expected transfer to the household that owns the respective bank, where a moral hazard problem constrains the bank's ability to obtain external funds, as in [Gertler and Karadi \(2011\)](#). The moral hazard problem gives rise to an endogenous leverage constraint: for given capital, total assets have to be consistent with that leverage constraint if any external funding is to be raised (i.e. the banker's optimization problem is subject to an incentive compatibility constraint).

Total assets of intermediary j at the end of period t are given by

$$p_{j,t}^B = q_t s_{j,t}^k + s_{j,t}^b, \quad (2)$$

where $s_{j,t}^k$ denote claims on intermediate goods firms by bank j that have the relative price q_t and that pay a net real return r_{t+1}^k at the beginning of period $t+1$, and $s_{j,t}^b$ are bank j 's government bond holdings that pay a net real return r_{t+1}^b at the beginning of period $t+1$. The balance sheet of bank j thus looks as follows:

$$p_{j,t}^B = d_{j,t}^B + n_{j,t}, \quad (3)$$

where $d_{j,t}^B$ denote deposits by households at intermediary j and $n_{j,t}$ denotes the intermediary's net worth. The latter evolves over time as the difference between earnings on assets and interest payments on liabilities:

$$n_{j,t+1} = (1 + r_{t+1}^k)q_t s_{j,t}^k + (1 + r_{t+1}^{b,B})s_{j,t}^b - (1 + r_{t+1}^d)d_{j,t}^B = (r_{t+1}^p - r_{t+1}^d)p_{j,t}^B + (1 + r_{t+1}^d)d_{j,t}^B, \quad (4)$$

where r_{t+1}^p is the net ex-post real portfolio return. Define portfolio weights $\omega_{j,t} = q_t s_{j,t}^k / p_{j,t}^B$ and $1 - \omega_{j,t} = s_{j,t}^b / p_{j,t}^B$, so r_t^p satisfies:

$$1 + r_t^p = (1 + r_t^k)\omega_{j,t-1} + (1 + r_t^{b,B})(1 - \omega_{j,t-1}). \quad (5)$$

At the beginning of period $t + 1$, after financial payouts have been made, an individual bank intermediary continues operating with probability θ and exits with probability $1 - \theta$, in which case it transfers its retained capital to its household. The bank manager's objective in period t is therefore to maximize the expected value of discounted terminal wealth:

$$V_{j,t} = E_t \sum_{i=0}^{\infty} (1 - \theta) \theta^i \beta^{t+1+i} \Lambda_{t,t+1+i} n_{j,t+1+i}. \quad (6)$$

For that he has to decide on size and allocation of his asset portfolio given his initial net worth $n_{j,t}$. Following [Gertler and Karadi \(2011\)](#), we assume that a costly enforcement problem constrains the ability of banks to obtain funds from depositors. In particular, at the beginning of period t , before financial payouts are made, the banker can divert a fraction λ^* of total current assets, $\lambda^* p_{j,t}^B$. If that happens, depositors will force the bank into bankruptcy to recover the remaining assets, $(1 - \lambda^*) p_{j,t}^B$, but it is too costly for the depositors to recover the funds that the banker diverted. Of course, in that case the banker will forfeit the continuation value, $V_{j,t}$. Accordingly, for the depositors to be willing to supply funds, the opportunity cost to the banker of diverting assets should not be smaller than the gain from diverting assets, i.e. the incentive constraint $V_{j,t} \geq \lambda^* p_{j,t}^B$ must be satisfied. So the maximization problem becomes:

$$\max_{s_{j,t}^k, s_{j,t}^b} V_{j,t} \quad \text{s.t.} \quad V_{j,t} \geq \lambda^* p_{j,t}^B. \quad (7)$$

The solution to this maximization problem closely follows [Gertler and Karadi \(2011\)](#) and is therefore relegated to the appendix. A key component of the solution is the leverage ratio ϕ_t which denotes the bank's leverage ratio of assets over net worth and limits the bank's leverage ratio to the point where the banker's incentive to cheat is exactly balanced by the cost.

3.4. Arbitrage and interest rates

So two different intermediaries hold government debt on their balance sheet, with only one (banks) subject to financial frictions, and the other (MMFs) not. But there is arbitrage on the side of consumers, who hold both debt (frictionlessly through MMFs) and bank deposits. As a consequence, return differentials on the asset side between bonds held by banks and by MMFs are unavoidable because of the frictions banks are subject to and the associated credit spreads. Then the only way to avoid corner solutions without banks holding any sovereign debt is to block (or otherwise assume the absence of) arbitrage between the pricing of bonds held by banks and of bonds held by MMFs. We do think there are convincing reasons for banks to hold on to underpriced debt in spite of the obvious consequence that they could profit from reselling the bonds into the market (if that is what one may call our frictionless intermediaries, i.e. the MMFs). First, the underpricing may be due to strong public preferences for placing bonds with banks rather than in the market; [Gennaioli et al. \(2014a\)](#) suggest that as one possible explanation for the patterns of bank holdings of sovereign debt they describe. [Ongena et al. \(2016\)](#) go even further and claim strong empirical evidence for precisely this driver of bank holdings of home country sovereign debt at the exclusion of a host of other explanations; admittedly their data cover a more limited time period and sample of countries than ([Gennaioli et al., 2014a](#))'s work does. [Horváth et al. \(2015\)](#) in a more recent paper report a mixture of home bias drivers, including public sector pressure as one rationale for which they find empirical evidence. To achieve that aim governments presumably have to offer larger discounts/higher returns when placing the bonds. Thus a future dollar sold to banks is cheaper to buy (raises less money now) than when the same promise to pay is placed in the direct capital market (our MMFs). In that sense bank held debt will be underpriced in bank portfolios, just as happens in our setup.

But this obviously leads to the next question: why do banks not sell these bonds for an easy profit? A rationale for why banks do not sell the bond into a segment of the market where they are higher priced is given in the corporate finance literature on why banks seemingly irrationally hold on to assets particularly when they are undercapitalized ([Diamond and Rajan, 2011](#)): because of limited liability, bank managers/shareholders only consider the upper part of the return distribution (ignore the left tail) and therefore fundamentally value contingent return assets higher than they are valued in the market where valuation is based on the entire return distribution. So they refuse to sell assets although they could seemingly and objectively profit from doing so. This is a particularly strong argument when banks are undercapitalized, a point stressed by [Diamond and Rajan \(2011\)](#), i.e. precisely the circumstances we think are most relevant for the issues analyzed in our paper. An alternative driver could well be regulatory distortions. The regulatory distortion argument is that capital requirements are assessed against risk-weighted assets, and sovereign debt everywhere in the world has a zero risk weight. If raising equity is costly, as is widely assumed in the literature on financial intermediation,³ banks have a strong incentive to

³ See, for example, [VanHoose \(2007\)](#) for this point in relation to capital requirements.

hold on to more sovereign debt than portfolio considerations would justify (Korte and Steffen (2015) argue this although Ongena et al. (2016) do not find support for that explanation, at least not for the European Debt Crisis period). Costly equity in combination with a binding capital-to-asset ratio constraint and zero risk weights on sovereign debt would then also explain both why banks would excessively buy into sovereign debt and why banks would forgo subsequent arbitrage opportunities. Yet another argument for banks valuing sovereign debt more than portfolio considerations alone would justify is given in Balloch (2016), who argues that banks have a bias towards sovereign debt because of its value as collateral in the secured funding market.

To close the model, the monetary authority is assumed to set the risk-free nominal interest rate on deposits r_t^n to stabilize inflation and output according to a Taylor rule of the form

$$r_t^n = (1 - \rho_r)[r^n + \kappa_\pi(\pi_t - \bar{\pi}) + \kappa_y \log(y_t/y_{t-1})] + \rho_r r_{t-1}^n + \varepsilon_{r,t}, \quad (8)$$

with $\kappa_y \geq 0$, $\kappa_\pi > 1$, $\rho_r \in [0, 1)$ and $\varepsilon_{r,t} \sim N(0, \sigma_r^2)$. The parameter $\bar{\pi} \geq 1$ stands for the inflation target. The strength of the monetary authority's reaction to fluctuations of inflation and output is determined by the parameters κ_π and κ_y , where we have imposed the Taylor principle as $\kappa_\pi > 1$ (Taylor, 1993). We also allow for an interest rate smoothing component in the Taylor rule, where the strength of interest rate smoothing is controlled by the parameter ρ_r .⁴ The following Fisher relation then defines the ex-post gross real interest rate on deposits:

$$1 + r_t^d = (1 + r_{t-1}^n) \pi_t^{-1}. \quad (9)$$

Notice that the model emphasizes the direct links of e.g. central bank lending rates to intermediary funding rates by the choice of the deposit rate as the instrument for monetary policy. The interest rate on government bonds is however endogenously determined in the general equilibrium. As noted above, the interest rate of MMFs is:

$$r_t^{b,MMF} = r_t^d. \quad (10)$$

3.5. Fiscal policy

The government conducts purchases of the final good and financial sector policies.

3.5.1. Government purchases in response to a financial crisis

Government purchases g_t consist of a stochastic part $\tilde{g}_t$ plus a possible response to shocks to the capital quality ξ_t :

$$g_t = \tilde{g}_t + \varsigma(\xi_{t-l} - \bar{\xi}), \quad (11)$$

with $\varsigma \leq 0$, $l \geq 0$ and $\bar{\xi} > 0$.⁵ The variable $\tilde{g}_t$ follows an autoregressive process in logs. We introduce concurrent shocks that come as a surprise, $\varepsilon_{g,t}^u$, and shocks that are pre-announced four periods ahead of time, $\varepsilon_{g,t-4}^a$, which results in the following process for government expenditure: $\log(\tilde{g}_t/\bar{g}) = \rho_g \log(\tilde{g}_{t-1}/\bar{g}) + \varepsilon_{g,t}^u + \varepsilon_{g,t-4}^a$, with $\varepsilon_{g,t}^x \sim N(0, \sigma_{g,x}^2)$ for $x = u, a$, $\rho_g \in [0, 1)$, and $\bar{g} > 0$. The parameter ς determines the spending response to shocks to the quality of capital. Below, this shock serves as the initiating disturbance leading to a financial crisis. If $\varsigma < 0$, spending increases during the crisis above its steady state value. If $\varsigma = 0$, there is no government intervention. Through the parameter l , the response occurs contemporaneously ($l=0$) or with some lag ($l > 0$). Although it may seem less appealing from a practical point of view than, for instance, an endogenous output feedback, an exogenous feedback of this type makes the policy experiments conducted below comparable by excluding second-round effects of induced output changes back into more government interventions.

3.5.2. Government interventions in the financial sector

We also allow for government interventions in the financial sector. In particular, the government is willing to provide funds $n_{g,t}$ to financial intermediaries according to a rule that is symmetric to the spending rule:

$$n_{g,t} = \varkappa(\xi_{t-l} - \bar{\xi}), \quad (12)$$

with $\varkappa \leq 0$ and $l \geq 0$. According to this rule, if $\varkappa < 0$, the government provides funds in the face of shocks to the quality of capital, which can again occur contemporaneously or with some lag as determined by l . If $\varkappa = 0$, there are no government interventions in the financial sector.

3.5.3. Government financing through the financial sector

Let b_{t-1} (b_t) denote the stock of government debt at the beginning (at the end) of period t . Taxes follow the rule

$$\tau_t = \bar{\tau} + \kappa_b(b_{t-1} - b), \quad (13)$$

with $\kappa_b > 0$ and $\bar{\tau} > 0$. This tax rule ensures fiscal solvency for any finite initial level of debt (Bohn, 1998). The introduction of small distortionary taxes for debt repayment is unlikely to change our main qualitative conclusions.

⁴ The specification of the Taylor rule uses the log deviation of current output from last period's output, to approximate the output gap that appears in the original version of Taylor (1993), following common specifications in empirical macroeconomic models (e.g., Christoffel et al., 2008).

⁵ Variables without time subscript denote non-stochastic steady state values.

Further, the government allocates a share Δ_t of new bonds issued to finance primary deficits to banks, and the remaining share is allocated to MMFs.⁶ The stock of government debt held by banks therefore satisfies the following law of motion:

$$s_t^b = \Delta_t(g_t - \tau_t + n_{g,t} - \tilde{n}_{g,t}) + (1 + r_t^{b,B})s_{t-1}^b. \quad (14)$$

Similarly, the stock of debt held by MMFs evolves in the following way:

$$d_t^{MMF} = (1 - \Delta_t)(g_t - \tau_t + n_{g,t} - \tilde{n}_{g,t}) + (1 + r_t^{b,MMF})d_{t-1}^{MMF}. \quad (15)$$

We assume for simplicity that $\Delta_t = \Delta$ for all t .

3.6. Aggregation and market clearing

Finally, individual supply and demand equations can be aggregated to derive aggregate commodity and financial market clearing conditions, and define the equilibrium sequences that satisfy the latter and the various first-order and transversality conditions stemming from the optimization by firms, households and financial intermediaries. The details are once again in the appendix.

4. Model analysis

In this section the model is used to analyze the effects of deficit-financed changes in government purchases, as well as alternative fiscal policy responses to a simulated crisis, focusing also on the effectiveness of fiscal policy at the Zero Lower Bound. Before showing the results, basic business cycle statistics from the calibrated model are presented. The aim of this exercise is to convince the reader that the model can be taken seriously as a laboratory to study fiscal multipliers.

4.1. Calibration

Table 1 lists the choice of parameters for the baseline version of the model. For comparability with the existing academic literature we have chosen parameters that are commonly used in similar DSGE models.⁷ In particular, the calibration follows Gertler and Karadi (2011). This concerns the subjective discount factor β , the degree of habit formation ν , the inverse Frisch elasticity of labor supply φ , the elasticity of substitution among intermediate goods ϵ , the Calvo probability of keeping prices fixed ψ , the capital share in production α , the investment adjustment cost parameter γ , and the depreciation rate δ . The parameters in the monetary policy rule are set to conventional values. In addition, the debt feedback on taxes κ_b is set to a small value but large enough to have stability conditions satisfied in both the version of the model with financial frictions and the version without financial frictions and financial intermediaries (see the appendix).⁸

Following again Gertler and Karadi (2011), the steady state leverage ratio ϕ is set to four to roughly match aggregate U.S. financial data. The steady state credit spread Γ is set to one hundred annual basis points to match the pre-2007 spreads of bank lending rates to risk-free bonds. The average survival rate of bankers $\theta = 1/(1 - \theta)$ is set to sixteen quarters (thus smaller than in Gertler and Karadi, 2011) by calibrating the survival probability θ , to make sure that the proportional transfer to entering bankers χ is positive (see the appendix). To roughly match U.S. macroeconomic data, the steady state ratio of government spending over GDP g/y is set to 20% and the ratio b/y is set to 2.4, which implies an annual debt-to-GDP ratio of 60%.

4.2. Matching business cycle statistics

To assess whether the calibrated model can mimic the broad dynamics in the data, this section compares basic business cycle statistics implied by the model with pre-2008 U.S. data.⁹ For this exercise, the parameters of the three economic shocks in the model, i.e. the shocks to TFP, government expenditures and the monetary policy rate, are chosen to match empirical moments for some variables as closely as possible.¹⁰ In particular, the parameters of the productivity shock and the government spending shock (parameters ρ_a , ρ_g , σ_a , and $\sigma_{g,u}$) and the standard deviation of the i.i.d. shock to the monetary policy rule (parameter σ_r) are chosen to match the standard deviations of the cyclical components of U.S. real GDP, real government expenditure on consumption and investment, the Federal funds rate, commercial banks' credit, and the

⁶ Note that we only consider one-period government debt so the stock of debt is rolled over each period and the allocation of debt over banks and MMFs can be changed in each period.

⁷ A large fraction of those parameters is based on attempts to match moments of U.S. data. We do not want to imply that the mechanisms we discuss are particularly relevant to the U.S., given the relatively small role banks play in U.S. debt markets. But the interaction between bank balance sheets and sovereign debt has been at the core of the euro area crisis and is also relevant in most emerging markets.

⁸ In the case when $\Delta = 1$ with full deficit financing through banks, we set $\kappa_b = 0$.

⁹ We use pre-crisis data since we do not believe that post-crisis data is generated by the same process.

¹⁰ Formally, let θ be the $k \times 1$ vector of parameters, y_t a vector of variables at time t for T observations, $\pi(y^T)$ an $m \times 1$ vector of empirical moments to be matched where y^T denotes the whole sample, and $f(\theta)$ their model counterparts. The estimation consists of minimizing $g_T(\theta)' W g_T(\theta)$, where $g_T(\theta) = \pi(y^T) - f(\theta)$ and W is an $m \times m$ matrix. For our exercise, $m = k$ and W is the identity matrix.

Table 1
Model parameters and steady state values.

Parameter	Value	Definition
<i>Households</i>		
β	0.990	Subjective discount factor
ν	0.815	Degree of habit formation
φ	0.276	Inverse Frisch elasticity of labor supply
<i>Financial intermediaries</i>		
λ^*	0.220	Fraction of assets that can be diverted
θ	0.938	Survival probability of bankers
χ	0.044	Proportional transfer to entering bankers
<i>Goods-producing firms</i>		
ϵ	4.167	Elasticity of substitution
ψ	0.779	Calvo probability of keeping prices fixed
α	0.330	Share of effective capital in production
<i>Capital-producing firms</i>		
δ	0.025	Depreciation rate of capital
γ	1.728	Investment adjustment cost parameter
<i>Policy</i>		
κ_b	0.020	Government debt feedback on taxes
ρ_r	0.800	Interest rate smoothing parameter
κ_π	1.500	Inflation feedback on nominal interest rate
κ_y	0.125	Output feedback on nominal interest rate
<i>Steady state values</i>		
ϕ	4.000	Banks' leverage ratio
Γ	0.003	Banks' credit spread
r^d	0.010	Households' return on deposits
$r^{b,MMF}$	0.010	MMFs' return on government bonds
$r^{b,B}$	0.013	Banks' return on government bonds
r^k	0.013	Banks' return on capital
r^e	0.020	Banks' return on equity
g/y	0.200	Government-spending-to-GDP ratio
b/y	2.400	Government-debt-to-GDP ratio

Note: The return on equity is calculated as $r^e = r^p \phi - r^d(\phi - 1)$, where $r^p = r^{b,B} = r^k$.

spread between Moody's Baa corporate bond yield and the Federal funds rate as a measure of the credit spread, for the period 1986Q1–2007Q4. For simplicity, only the unanticipated shock to government expenditures is activated for the purpose of this exercise. Simulated moments from the model for these and other variables are then compared to the data. Those other variables are quarterly CPI inflation, federal government debt, and commercial banks' equity capital.¹¹

The results of this exercise are documented in Table 2. The second and fifth columns of Table 2 display the point estimates of selected empirical moments (standard deviations and first-order autocorrelations) and associated standard errors in parentheses.¹² The remaining columns display the simulated moments from the model for two cases: $\Delta = 1$ (allocation of debt through banks) and $\Delta = 0$ (allocation of debt through MMFs). To convey the statistical significance of differences between the empirical and model moments, associated t -statistics are reported in square brackets. The standard deviations and autocorrelations of most variables are matched well by the model, as indicated by the t -statistics below 2 in absolute value associated with the moments of real GDP, government expenditure, government debt, the Federal funds rate, the credit spread, and bank credit. One exception is inflation, whose volatility and persistence are overstated by the model. In addition, while the autocorrelation of bank capital is matched well, its volatility is exaggerated. Interestingly, the model with $\Delta = 1$ shows a significantly better fit for the volatility of government debt than the model with $\Delta = 0$. Several other moments are also better matched by the model with $\Delta = 1$, with the exception of the Federal funds rate. We conclude from this exercise that the model provides an overall reasonable description of the basic properties of the dynamics in the data.

4.3. Effects of a surprise spending shock

The dynamics of the response to a surprise spending shock are examined first. Fig. 1 shows the responses of selected variables to an unanticipated increase in government spending (goods purchases) that is normalized to 1% of GDP on impact and that is persistent with autocorrelation coefficient $\rho_g = 0.8$. We consider the case of full bank financing of the deficit, and

¹¹ All series were obtained from the St. Louis Fed's database at <http://research.stlouisfed.org/fred2>. The following transformations are made: nominal quantities are deflated by the CPI, real quantities are converted into natural logarithms while other variables are demeaned, and the cyclical components are estimated with the Hodrick–Prescott filter with a smoothing parameter of 1600. The moments from the model are computed based on 10,000 simulations parting from the steady state. The model variables are measured in the same way as in the data, including the use of HP-filtered series.

¹² The standard errors were obtained using the Delta method and GMM.

Table 2
Moment comparison.

Variable	Standard deviations			Autocorrelations		
	Data	Model		Data	Model	
		$\Delta = 1$	$\Delta = 0$		$\Delta = 1$	$\Delta = 0$
GDP	0.91 (0.11)	0.96 [−0.43]	0.95 [−0.38]	0.87 (0.22)	0.90 [−0.15]	0.91 [−0.16]
Government expenditure	0.87 (0.07)	0.86 [0.02]	0.87 [0.01]	0.64 (0.15)	0.54 [0.63]	0.63 [0.05]
Government debt	1.70 (0.28)	1.51 [0.67]	1.03 [2.39]	0.83 (0.31)	0.59 [0.76]	0.87 [−0.14]
CPI inflation	0.33 (0.04)	0.43 [−2.90]	0.46 [−3.68]	0.04 (0.10)	0.45 [−4.23]	0.47 [−4.48]
Federal funds rate	1.09 (0.11)	0.87 [2.10]	0.97 [1.16]	0.92 (0.19)	0.81 [0.58]	0.82 [0.51]
Credit spread	1.00 (0.10)	1.06 [−0.61]	1.08 [−0.89]	0.89 (0.18)	0.62 [1.45]	0.64 [1.37]
Bank credit	1.31 (0.11)	1.25 [0.51]	1.22 [0.82]	0.78 (0.16)	0.58 [1.26]	0.55 [1.40]
Bank capital	2.22 (0.24)	4.81 [−10.77]	4.93 [−11.27]	0.71 (0.17)	0.58 [0.73]	0.60 [0.67]

Note: Standard errors in round brackets, *t*-statistics in square brackets.

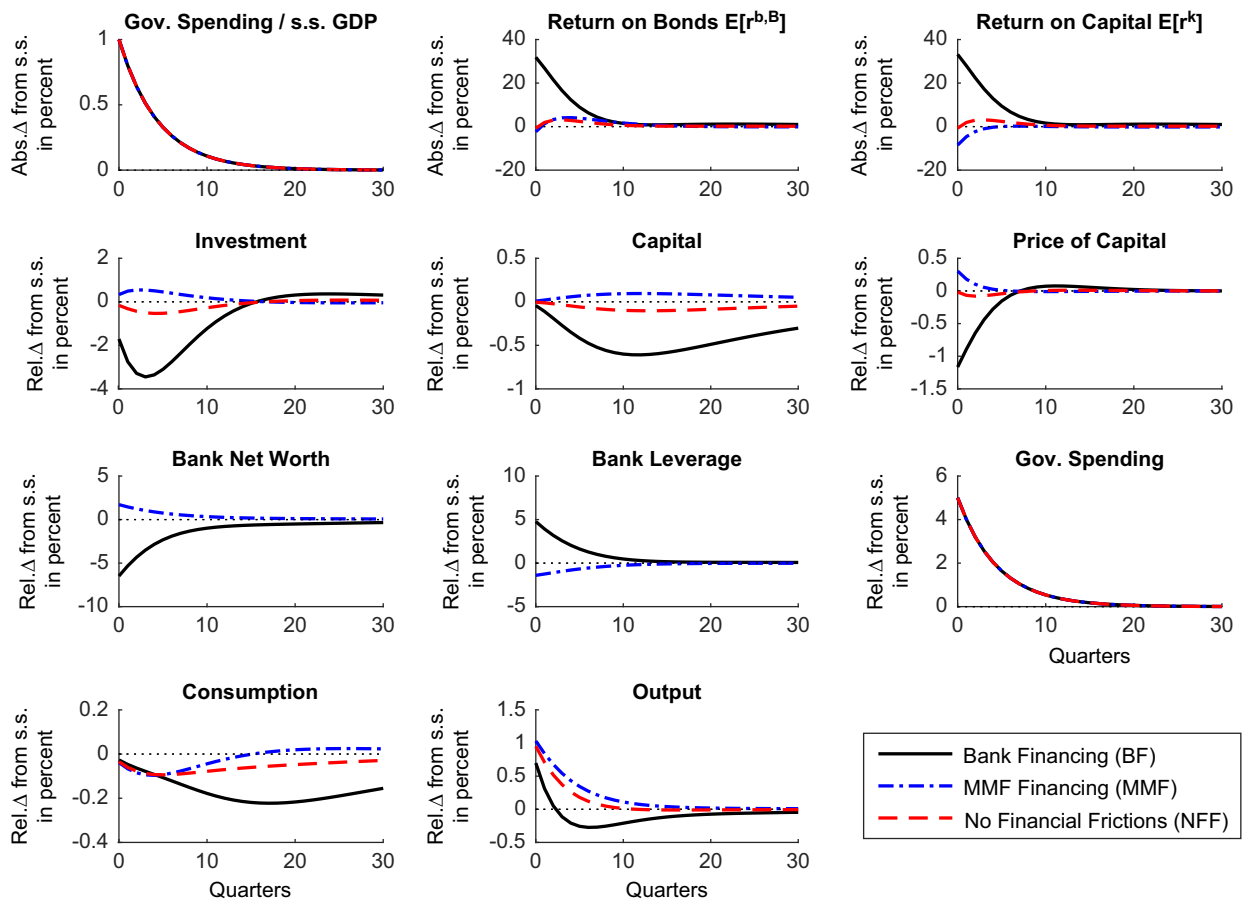


Fig. 1. Impulse responses to a surprise spending shock. Note: Unexpected increase in government spending in quarter 0 (innovation $\varepsilon_{g,t}^u$) by 1% of GDP relative to steady state. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

the case of full household financing of government debt, i.e. when all public debt is debt placed through frictionless MMFs.¹³ The figure also shows the impulse responses from a version of the model without financial intermediaries at all, eliminating financial frictions altogether, as another reference case.¹⁴ This version of the model is described in the appendix. Comparing the Bank Financing (BF) case with the MMF Financing where government debt is not subject to agency problems (MMF) shows the impact of placing debt at already financially distressed banks; comparison with the No Financial Frictions (NFF) case shows the impact of financial fragility per se, since there are no leverage constraints in the NFF case.

In the NFF model without intermediaries, the spending expansion raises output one-for-one on impact and afterward the output effect decreases with the government spending impulse. However, in the model with bank financing of the spending expansion, the output response is first of all much smaller on impact: the impact multiplier falls to substantially below 1.0. And second, the output response tails off much more quickly; after about a year the within-period multiplier even turns negative and stays so. So crowding out effects are immediate and quite persistent.

Underlying those effects are the funding pressures that are put on banks by a deficit-financed fiscal expansion. The fiscal expansion raises both expected interest rates through the associated tightening of bank balance sheet constraints and intermediary balance sheet adjustments. The precise mechanism through which this occurs is explained in more detail in Section 4.4 below. As a consequence of the rise in borrowing costs, the demand for capital by intermediate goods firms and thus investment by capital producers is crowded out: in the BF case investment goes down instead of going up as it does in the MMF and NFF cases. The fall in investment is amplified by the financial accelerator mechanism, like in Gertler and Karadi (2011): procyclical variation in intermediary balance sheets amplifies the negative effects. Falling investment leads to a falling price of capital, which lowers intermediary net worth and thus further tightens intermediary constraints, which further raises borrowing costs such that investment falls by more, further decreasing asset prices, and so forth. These effects feed through the whole economy as falling wages discourage household labor supply and as the associated tightening of the households' budget constraints depresses consumption.

Interestingly enough, the spending expansion would be more effective if it could be financed by households directly, with the government bypassing leverage-constrained financial intermediaries, but with financial frictions in place for private financing needs. This can be seen by comparing the MMF case (in effect no financial frictions for public debt but frictions for private financing needs) with the NFF case where no financial frictions are present at all. Under household financing, the intermediary balance sheet mechanism makes the spending expansion comparatively more effective as the build-up of investment (which occurs under household financing just like in the model without intermediaries) raises the price of capital over time, which in turn eases intermediary balance sheet constraints. As a consequence, an analysis of stimulus policies in a model like Gertler and Karadi (2011), where banks face no leverage constraint on financing acquisitions of public debt but do face such constraints on the financing of claims on the private sector, is likely to lead to the conclusion that the benefits of such policies are *enhanced* by the relevant financial frictions. The latter is the exact opposite of our results: in our BF case, financial frictions *reduce* multipliers, eventually even reversing their sign and even turning negative.

Fig. 2 highlights that point: there we let the fraction of debt allocated to leverage-constrained banks (with the rest going to unconstrained MMFs) slide from $\Delta \rightarrow 0$ (all debt goes through MMFs) to $\Delta \rightarrow 1$ (all debt goes through banks). Without intermediation by leverage-constrained intermediaries, multipliers increase substantially, both on impact (the upper panel in Fig. 2) and cumulative. Cumulative multipliers are defined as the sum of all output effects divided by the sum of all government spending increases. Impact multipliers fall below one as Δ increases, as we already saw in Fig. 2; a crowding out effect sets in immediately once the share of debt finance through leverage-constrained intermediaries goes above approximately 40%. And cumulative multipliers, which are anyhow lower than one as is usually the case in New Keynesian models, fall by about a third as financing goes through leverage-constrained banks entirely.

4.4. Effects of a pre-announced spending shock

The intermediary balance sheet adjustments in interaction with leverage constraints that are at the heart of our model become very clear when we look at the effects of a spending increase that is pre-announced one year in advance. This experiment allows to distinguish the relevant expectational effects from other more direct channels since the direct impact of the spending increase arrives a full four quarters after the announcement. Everything that happens before the actual increase takes place is uniquely driven by expectational effects. The example should also be of substantial practical interest: moving from budgetary plans to implementation takes about a year in most countries.

Fig. 3 shows the response of the economy to news in quarter 0 that spending is going to increase by 1% of GDP in quarter 4. The results show that with intermediary financing of public debt, output falls immediately after the announcement of the spending expansion. The diagrams also highlight the relevant channel: a substantial decline in investment in anticipation of the future tightening that deficit financing will bring takes place when government debt is subject to agency frictions, i.e. is intermediated through the banking system. Under direct household financing bypassing the banking sector in favor of MMFs, the spending expansion announcement would have almost no effect on output after the news but before the implementation.

¹³ This variant is observationally equivalent to one where financial intermediaries would take deposits from households but face no leverage constraints on their financing of public debt, like in Gertler et al. (2012).

¹⁴ In this variant of the model, borrowing rates co-move through arbitrage behavior by households that choose between bonds and deposits, and the central bank sets the interest rate on bonds.

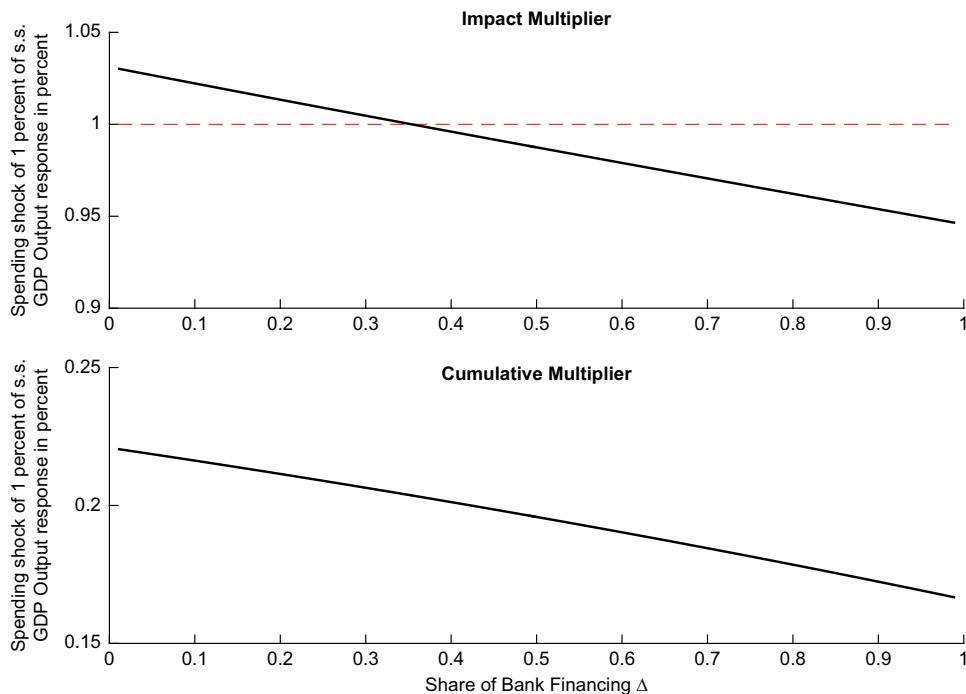


Fig. 2. Spending multipliers for different shares of bank financing Δ . *Note:* Impact multiplier is initial response of output to an unexpected increase of government spending by 1% of GDP relative to steady state. Cumulative multiplier is sum of output responses divided by sum of spending shocks over 1000 periods.

The underlying mechanism relies on two effects. A first effect works through a tightening of intermediary balance sheet constraints. The expected future increase in government primary deficits due to the upcoming spending expansion implies higher expected growth rates of bonds and total assets. The intermediaries have an incentive to accumulate assets due to a rise in the expected discounted marginal gain of assets. This incentive, however, tightens leverage constraints as indicated by a strong rise in the associated Lagrangian multiplier. The latter restricts the banks' asset demand and raises the costs of credit to both the government and intermediate goods firms during the announcement period (i.e. it raises credit spreads).

A second effect works through intermediary portfolio adjustments. Rising spreads discourage investment and lower the price of capital. Everything else equal, the fall in the price of capital enhances the rise in the expected return on capital, as the price of capital recovers again after the initial fall. Intermediaries thus shift their portfolios into assets with higher expected returns, i.e. claims on intermediate goods firms. The associated fall in the demand for bonds reduces the implicit bond price and raises the ex-ante nominal interest rate on bonds. The expected real rate on bonds increases, which adds to the rise in the expected overall portfolio return. The increase in the expected portfolio return further enhances incentives to accumulate assets and thus reinforces the first effect discussed above. Notice that the fall in the demand for bonds is only consistent with an equilibrium if the *ex-post* real interest rate on bonds falls given a fixed initial supply of bonds by the government. Hence, a planned fiscal expansion in this environment might give an impression of further fiscal space due to low interest rates when, in fact, there is none. Once the expansion takes place output shoots up to give the arguably misleading impression that it is actually effective, albeit only for a short time.

4.5. Policy responses to the financial crisis: fiscal stimulus and bank recapitalization

This section analyzes the effects of government policy interventions during a simulated financial crisis. The financial crisis itself is simulated through a negative shock to the capital quality parameter ξ_t , like in Gertler and Karadi (2011), by 5% on impact with autocorrelation coefficient $\rho_\xi = 0.5$. This shock triggers a response similar in terms of type, magnitude, and duration to the recent crisis. Other initiating shocks are conceivable but the specific type of shock is irrelevant for the qualitative implications of the analysis that follows. Also, and following Gertler and Karadi (2011), we assume that the monetary authority reduces its tendency to smooth interest rates in the face of the shock, to capture the notion that monetary policy tends to be more aggressive during a financial crisis.¹⁵ As seen in the figures below, absent any government intervention (the black solid lines), the deterioration in intermediary asset quality produces a sharp recession with a peak

¹⁵ The smoothing parameter ρ_r in the Taylor rule is reduced by half but not more so that non-negativity constraints on nominal interest rates remain satisfied.

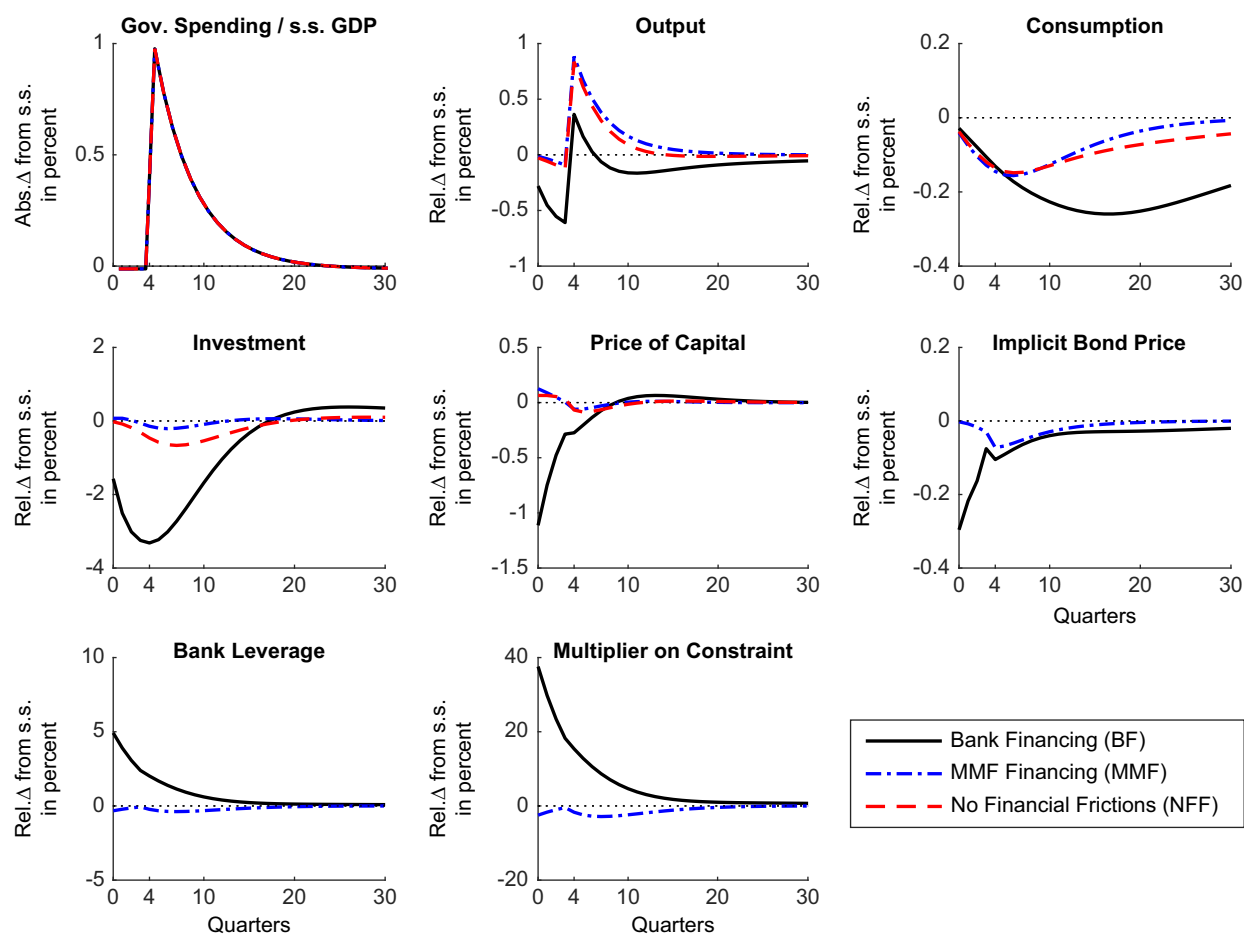


Fig. 3. Impulse responses to a pre-announced spending shock. *Note:* Announcement in quarter 0 (innovation $\varepsilon_{g,t}^a$) that government spending is going to increase by 1% of GDP relative to steady state in quarter 4. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

output decline of more than 4%, as intermediary net worth drops and credit tightens, leading to a sharp rise of the credit spread. Investment initially drops sharply and takes more than two years to recover; as a consequence, output takes more than four years to reach pre-shock levels again.

Consider first a fiscal stimulus in response to the financial crisis, but like in the previous section, with its disbursements four quarters delayed, reflecting the reality of budget processes. Fig. 4 illustrates the effects of such a countercyclical, persistent spending stimulus of 2% of GDP that is implemented with one year delay after the shock hits when added to the capital quality shock used to represent the financial crisis. The policy is implemented with a delay of one year after the initial shock occurs, but fully anticipated since it is a public budgeting process that is cause of the delay: this mimics that the spending package becomes known when approved but then requires time before it can actually be implemented. With this added delay, the deficit-financed stimulus is actually counterproductive: the fall in output is amplified. The peak decline in output reaches around 5%. The reason is that credit tightens immediately in the face of the upcoming spending expansion, as discussed in Section 4.4. Private investment thus falls immediately as the crisis hits, but the actual stimulus arrives later which explains why the output fall gets worse. Because of the anticipated increase in credit spreads associated with the upcoming fiscal deficits, the initial decline in output is further amplified instead of partially offset. The fall in investment and a larger decline in consumption (due to a higher future tax burden) eventually more than offset the initial output gain from the additional government spending, a short blip to the contrary during the initial implementation quarters notwithstanding. Notice again that in both cases the stimuli would be more effective if the leverage constraint can be bypassed, as in the benchmark MMF (household financing) case. The reason for this result is similar to the discussion in Sections 4.3 and 4.4, in the MMF case credit conditions do not tighten and private sector investment is not crowded out.

But if the financial crisis led to a deterioration of financial sector balance sheets, and that deterioration has reduced effectiveness of fiscal policy, a logical complementary step is to recapitalize the banking system simultaneously to reduce financial fragility and diminish its negative macroeconomic impact. Such steps have in fact been taken in all or most countries in Western Europe as well as in the U.S., albeit with a great variety in speed and methods chosen. In particular the

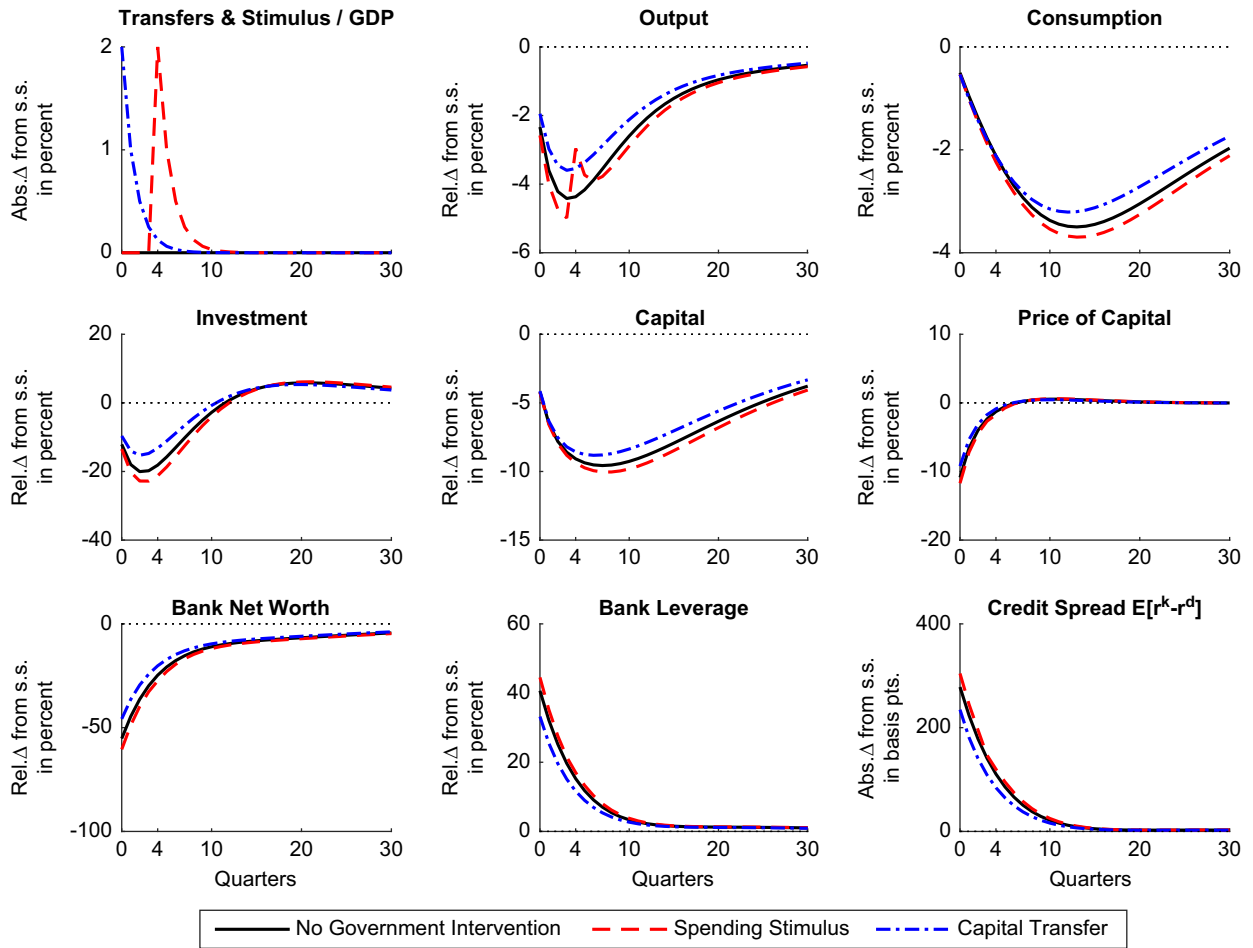


Fig. 4. Policy responses to a capital quality shock: effects of spending stimulus vs. bank recapitalization. *Note:* Initiating shock is unexpected decline in the quality of capital by 5% relative to steady state in quarter 0 with autocorrelation coefficient $\rho_\varepsilon = 0.5$. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

U.S. has acted substantially more decisively than authorities in Western Europe: In the U.S., first through the establishment of TARP in 2008 and later through the Supplementary Capital Assessment Program implemented in the summer of 2009, the major U.S. banks were forced to accept substantial balance sheet strengthening with very little time granted to reach that goal. In Western Europe banks were granted a five year horizon to bring themselves up to the higher Basel-III standard. The blue line in Fig. 4 shows the impact of a one off transfer to the bank intermediaries raising their capital ratios. Although the 2% of GDP chosen does not come close to offsetting the impact effect of the financial crisis shock, it does lead to significant macroeconomic improvements, in spite of the fact that the debt issued to finance the transfer itself adds to financial pressure. The recessionary impact of the financial crisis is substantially weakened by the capital infusion. Transfers, hypothetically perhaps, can bring stabilization gains even when deficit financing is fully subject to agency frictions. It follows immediately from Fig. 4 that the recession could also be dampened by a combined policy of spending stimulus and bank recapitalization.

4.6. Policy responses to the financial crisis: fiscal policy at the zero lower bound

A second set of policy measures pursued with particular vigor in the U.S. and somewhat more hesitantly in Western Europe was a push to radically lower interest rates to their Zero Lower Bound (ZLB). In the U.S. (and more recently also in the Eurozone) monetary expansion was continued even beyond that point in what is now referred to as Quantitative Easing, three major programs under which the Fed intervened aggressively in government debt markets (QE1 and QE3) and in commercial capital markets (QE2). See for an analysis of that experience [Gertler and Karadi \(2013\)](#), who use a model closely related to the one set up in this paper to analyze the output effects of these aggressive monetary policy packages. The question that is asked here is different, i.e. whether a continued policy of QE pushing the interest rate to its ZLB has an impact on the effectiveness of fiscal policy.

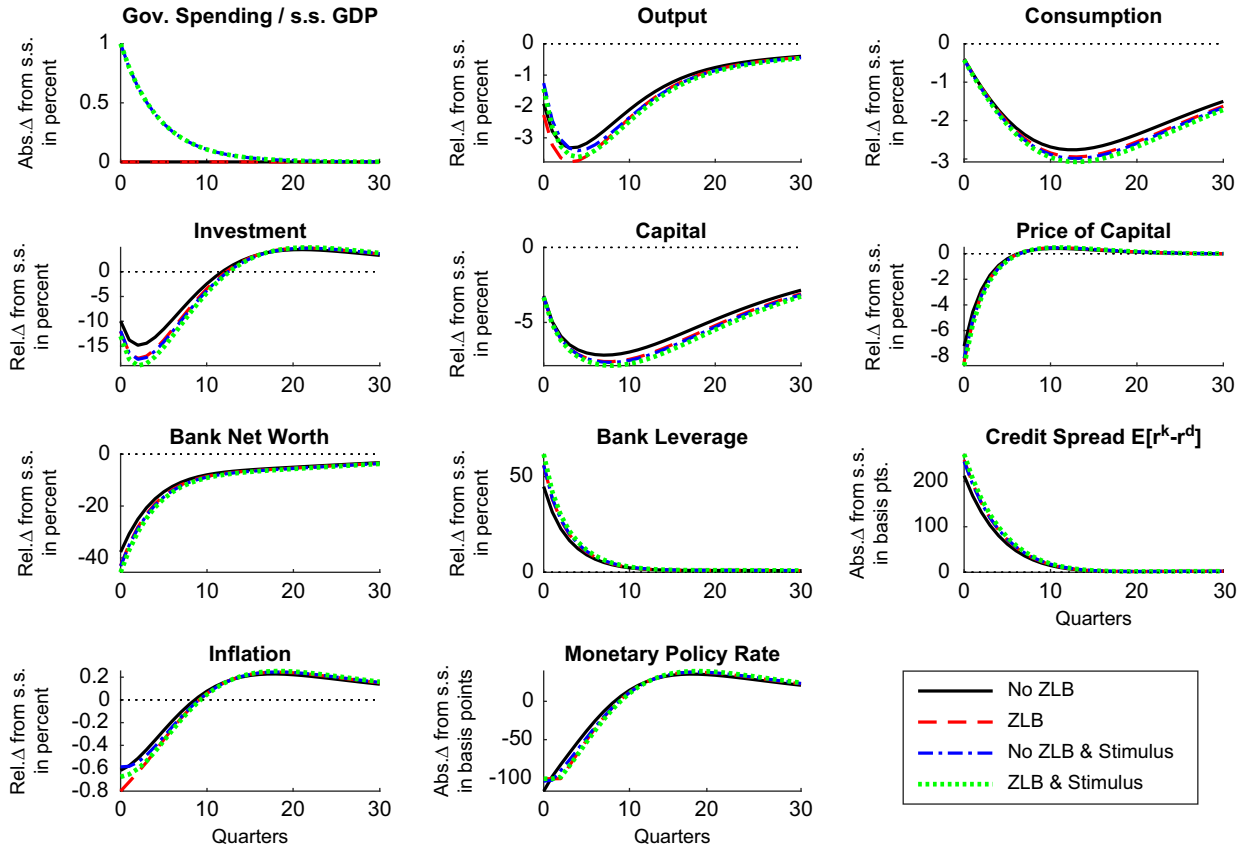


Fig. 5. Effects of a capital quality shock and spending shock at the ZLB. *Note:* Initiating shock is unexpected decline in the quality of capital in quarter 0 with autocorrelation coefficient $\rho_\varepsilon = 0.5$, of size such that the ZLB binds during 3 periods. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

Previous studies have shown that government spending expansions are more effective in stimulating output under the ZLB since the increase in government demand counteracts the deflationary spiral associated with the zero-bound state (e.g., [Christiano et al., 2011](#)). This channel is also present in our model, but we are more interested in analyzing the quantitative effects of a spending expansion under the ZLB when it is financed through leverage-constrained banks. We therefore compute a perfect foresight path of the economy with an inequality constraint on the interest rate: in each period the interest rate equals the maximum of the Taylor-rule-implied rate and zero after an initial capital quality shock. To make sure the economy actually does hit the ZLB after that capital quality shock, the interest rate smoothing parameter of the monetary policy rule is set to zero. That more aggressive policy stance is sufficient to drive the economy into the ZLB constraint during three quarters in the absence of the spending shock. The results are shown in [Fig. 5](#), which compares the different cases of not imposing the ZLB constraint, with the ZLB constraint and both with and without the spending shock. The latter is calibrated to 1% of steady state GDP with autocorrelation coefficient $\rho_g = 0.8$. The capital quality shock has significantly stronger effects when the ZLB binds, due to the rise in the real interest rate caused by the fall in inflation with the nominal interest stuck at the ZLB, which leads to an increase in desired saving that causes an even bigger fall in output. The deficit-financed spending stimulus seems to be successful in lifting the economy out of the ZLB state in the initial phase; however, the eventual output drop is larger than without the intervention and it is also larger than under a non-binding ZLB. These results are again due to the crowding out of private credit and investment when deficit financing is fully subject to agency frictions.

The last point is further documented in [Table 3](#), which displays the (impact and cumulative) spending multipliers with and without the ZLB, when the increase in government deficits is fully financed through leverage-constrained banks ($\Delta = 1$) or in a frictionless way through MMFs ($\Delta = 0$). When it is financed through frictionless markets, the spending expansion is found to be significantly more effective when the ZLB binds, with an impact multiplier larger than one, compared to the case where the ZLB does not bind, where the multiplier is smaller than one. However, when the spending expansion is financed through leverage-constrained banks then it is much less effective and the impact multiplier stays below one even if the ZLB binds. Interestingly, the cumulative multiplier is close to zero in this case meaning that, in terms of the effectiveness of fiscal policy, the costs due to deficit financing through leverage-constrained banks exactly compensate the gains due to the ZLB.

Table 3
Spending multipliers at the ZLB.

Multiplier	ZLB		No ZLB	
	$\Delta = 1$	$\Delta = 0$	$\Delta = 1$	$\Delta = 0$
Impact	0.84	1.28	0.66	0.91
Cumulative	0.00	0.43	– 0.14	0.16

Note: See Figs. 2 and 5.

Given the above results, what is the marginal improvement of the effectiveness of government spending due to ZLB in the two cases ($\Delta = 0$ and $\Delta = 1$)? To answer this question we compute the percentage increase in the multiplier due to the ZLB being active. When $\Delta = 0$, the impact multiplier increases by about 40% while the cumulative multiplier increases by more than 160%. However, when $\Delta = 1$ the impact multiplier increases by only 27% while the cumulative one increases by only 0.14 in absolute terms (against 0.27 when $\Delta = 0$). Hence, the increase in the effectiveness of government spending due to the ZLB is much smaller with frictions in debt intermediation in the short term while it is almost negligible over longer horizons.

5. Conclusion

After a wave of calls for fiscal stimulus to lift distressed economies out of the recession induced by the global financial crisis, economic growth did pick up after the implementation of large fiscal stimulus packages in the U.S., China and Western Europe. Many observers concluded that the fiscal stimuli were effective; see, for instance, [Romer \(2009\)](#) or [Romer and Bernstein \(2009\)](#), and also [CBO \(2009\)](#), [IMF \(2009\)](#), and [OECD \(2009\)](#). However, despite reduced credit spreads and diminishing job losses, the recovery turned out to be less robust than originally hoped, also in the U.S. where, at least in absolute size, the largest package was adopted (see [Mankiw, 2009](#)). Doubts about the impact of the implemented policies have started to come up and many observers have asked why the stimulus was not more effective (e.g., [Adams and Gangnes, 2010](#)).

Our model provides one possible answer to this question. One important element of our answer is the fact that it takes time to implement announced measures: in the case of the U.S., it took more than one year between the first plans for fiscal stimulus (cf. [Summers, 2008](#)) and the enactment of the American Recovery and Reinvestment Act of 2009. Our results suggest that an announced but delayed stimulus of this type can appear to be effective once it occurs, lowering credit spreads and raising output growth. However, the *announcement* of the stimulus will *deepen* the crisis before the implementation of any measures has taken place. Moreover, our analysis suggests that after its direct demand effects are realized, a stimulus package will be followed by lower medium-term growth than without any government interventions.

At the core of our analysis are furthermore the interactions between frictions in the banking system and the placement of public sector debt. Bank holdings of public debt are very important in most emerging market economies and are playing an increasingly crucial role in the macroeconomics of the euro area. This paper extends the recent literature on the macro-economic impact of leverage constraints on banks (see, in particular [Gertler and Karadi, 2011](#); [Gertler et al., 2012](#)) by introducing portfolio choice for banks, who in our setup can hold both public debt and claims on the private corporate sector. This introduces a new crowding-out channel if leverage constraints apply to all their assets, and not just to the financing of claims on the private sector as in [Gertler and Karadi \(2011\)](#) and [Gertler et al. \(2012\)](#). This paper also explores the implications of fiscal policy when government debt can be placed directly at consumers through frictionless Money Management Funds, bypassing the banks and their agency problems. Not surprisingly, as a larger fraction of debt is issued to consumers directly, the negative effects of a weak capital base of banks on the effectiveness of fiscal policy diminish. The findings are that while the Zero Lower Bound does improve the effectiveness of deficit-based government spending expansions when there are no leverage constraints, it does not add much when there are leverage constraints.

Our analysis supports the notion that even a timely conceived but slowly implemented stimulus can have undesirable crowding-out effects up front that may in fact be large enough to eventually offset its desired impact altogether. This paper highlights a channel through which more than one-for-one crowding out may happen that is very different from the example on the unavoidability of distortionary taxation, another mechanism through which crowding out of government expenditure can occur that has been stressed in the literature.¹⁶

Although our results are based on a calibrated model rather than on direct econometric estimation, the results are in line with many empirical studies. Our model predicts stronger crowding-out effects of spending-based fiscal expansions on investment than most of the VAR studies we have discussed in [Section 2](#) but the qualitative predictions are similar. In addition, our results are very much in line with empirical results obtained by authors focusing on economies in financial distress, like those reported in [Mulas-Granados et al. \(2009\)](#) and [Ilzetzki et al. \(2013\)](#).

¹⁶ See, for instance, [Coenen et al. \(2012\)](#), or [Drautzburg and Uhlig \(2015\)](#).

If fiscal stimuli are less effective than commonly thought, what can governments do to stabilize the economy in times of financial distress? This paper demonstrates that financial sector policies can be effective tools and also shows that a government fiscal stimulus program has a larger impact on output when accompanied by a bank recapitalization.

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Appendix A. Supplementary data

Supplementary data associated with this paper can be found in the online version at <http://dx.doi.org/10.1016/j.jmoneco.2016.04.007>.

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